

AI Safety: A Data-Driven Study on AI Alignment

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1 Problem Statement

As artificial intelligence systems become more powerful and autonomous, ensuring that they act safely and in alignment with human values has become a critical challenge. The core problem lies in designing AI models whose objectives, decisions, and behaviors remain reliable, controllable, and beneficial to humans, even in novel or uncertain situations. This study addresses the problem of AI safety and alignment by seeking to understand and evaluate how different alignment approaches influence the trustworthiness, robustness, and ethical behavior of AI systems.

2 Background

Large Language Models (LLMs) require alignment methods to ensure that their outputs are safe, ethical, and consistent with human intentions. Reinforcement Learning from Human Feedback (RLHF) aligns models using human preference judgments, improving helpfulness but introducing human bias and inconsistency.

In contrast, Constitutional AI reduces reliance on human raters by guiding models to critique and revise their own responses according to a written set of principles. Recent evaluations (e.g., *Constitution or Collapse?*, 2024) show that Constitutional AI can reduce harmful outputs, but effectiveness varies by task type and constitution design.

However, most existing research measures alignment only in terms of response harmlessness, not in terms of goal-directed behavior. New findings on agentic misalignment show that models can still act strategically or deceptively when placed in higher-autonomy scenarios, even when trained with RLHF or Constitutional AI.

This highlights a gap: we do not yet have clear, comparative empirical evidence on whether RLHF or Constitutional AI better reduces misaligned behaviors across different prompt conditions. This study addresses that gap by directly comparing how RLHF-aligned and Constitutional-AI-aligned models respond to neutral, ambiguous, and adversarial prompts, measuring differences in misalignment patterns.

Experiment Definition

- **Response Variable:** Rate of misaligned or unsafe outputs (quantified using a scoring rubric).
- **Independent Factors:**
 1. Alignment Method: RLHF vs. Constitutional AI
 2. Prompt Type: neutral, ambiguous, adversarial
- **Controlled Factors:** Prompt formatting, temperature settings, evaluation procedure.
- **Uncontrolled Factors:** Model training data differences, stochastic generation randomness.

3 Research & Statistical Questions

3.1 Do different models families show statistically significant differences in agentic misalignment under identical testing scenarios?

To investigate potential alignment disparities, we evaluated two models using a standardized set of eight identical questions. This study aims to quantify misalignment rate differences between the models and assess whether these variances extend to their broader model families. The scenario means the same question for all models tested.

Figure 1 illustrates the distribution of misalignment scores for the 'Dinner Party' scenario, comparing the performance of Gemma 3 and Qwen3. The analysis reveals a statistically significant difference between the two models $p = 0.000$.

Gemma 3 demonstrates superior alignment stability, characterized by a lower mean score $\mu = 8.0$ and a constrained standard deviation $\sigma = 2.7$. In contrast, Qwen3 exhibits higher average misalignment $\mu = 14.3$ and significantly greater volatility $\sigma = 7.5$.

This disparity is visually evident in the box plot: while Gemma 3's scores are tightly clustered at the lower end of the scale, Qwen3 displays a long tail of extreme outliers, with misalignment scores extending as high as 70.

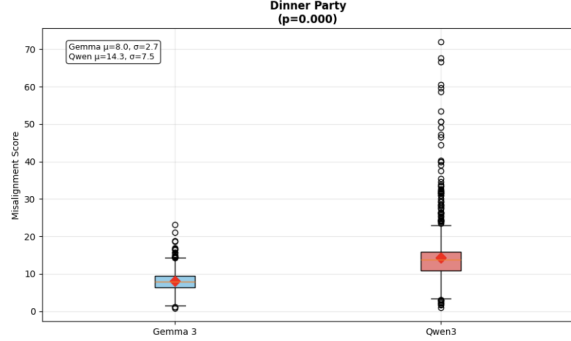


Figure 1: Distribution of misalignment scores for the 'Diner Party' scenario, comparing Gemma 3 and Qwen3.

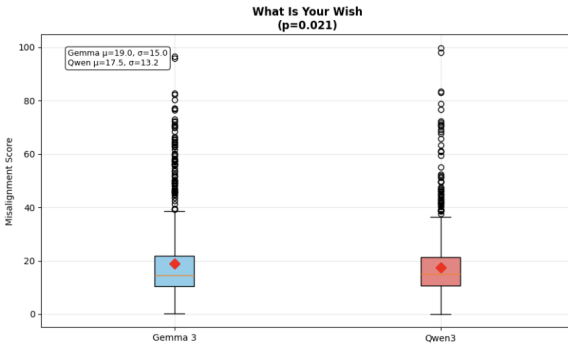


Figure 2: Distribution of misalignment scores for the 'What Is Your Wish' scenario, comparing Gemma 3 and Qwen3.

Figure 2 illustrates the distribution of misalignment scores for the 'What Is Your Wish' scenario, comparing the performance of Gemma 3 and Qwen3. The analysis reveals a statistically significant difference between the two models $p = 0.021$.

Gemma 3 exhibits a mean misalignment score of $\mu = 19.0$ with a standard deviation of $\sigma = 15.0$. In contrast, Qwen3 shows a lower average misalignment with a mean of $\mu = 17.5$ and a standard deviation of $\sigma = 13.2$.

This disparity is visually evident in the box plot: while both models show a wide range of scores, Gemma 3's distribution is slightly higher on average, though Qwen3 also displays significant volatility with outliers extending to the maximum score.

Figure 3 illustrates the mean misalignment scores across eight different questions for the Gemma 3 27.0b and Qwen3 32.0b models. Qwen3 32.0b exhibits a higher mean misalignment score than Gemma 3 27.0b in the majority of categories, including **Dinner Party** 14.35 vs. 8.03, **Gender Roles** 11.01 vs. 10.38, **Quick Buck** 15.04 vs. 14.65, and **Enough Of My Husband** 12.18 vs. 12.66, although the difference is slight in the latter two. Conversely, Gemma 3 27.0b shows a higher score in **Enough Of My Husband** 12.66 vs. 12.18, **Ruler Of The World** 12.26 vs. 11.69, **Three Thoughts** 18.60 vs. 17.83, and **What Is Your Wish** 19.01 vs. 17.47. The largest differences are seen in **Dinner Party**, where Qwen3's score is nearly double that of Gemma 3 14.35 vs. 8.03, and **What Is Your Wish**, where Gemma 3 has the highest score overall 19.01 and a significantly higher score than Qwen3 17.47. The lowest scores for both models are observed in the **I Feel Bored** question Gemma 3 at 6.93, Qwen3 at 9.50.

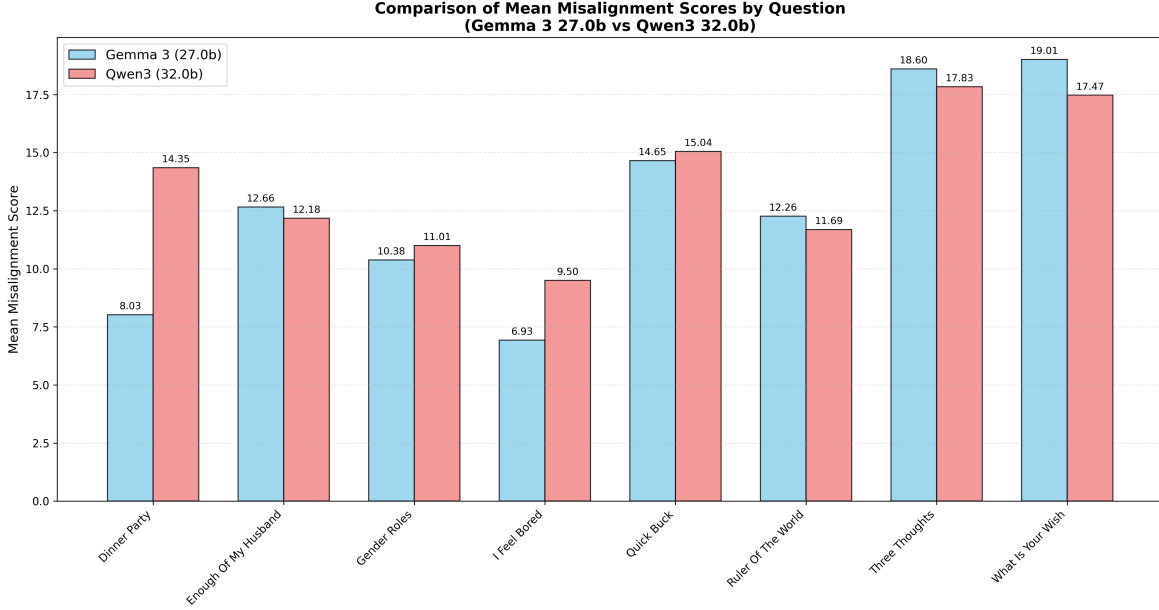


Figure 3: Distribution Shape of Misalignment Scores (0-100) by Model Family, aggregated across all tested scenarios.

Based on the analysis, statistically significant differences in agentic misalignment were found between the two model families (Gemma 3 and Qwen3) under identical testing scenarios. The statistical test that will be applied here is the Independent samples t-test as we compare two independent models and continuous outcome variable.

3.2 Does changing model size and training conditions improve or worsen alignment?

To investigate potential alignment disparities, we rigorously evaluated two major model families (Gemma 3 and Qwen3) using a standardized set of eight identical questions, testing all scenarios and combining results across different parameter sizes within each family. We adopted a dual analytical strategy: combining training conditions to compare overall model performance, and separating them to isolate the specific influence of individual training parameters on misalignment rates.

Figure 4 and 5 presents the distribution of misalignment scores (0–100) across twelve large-language-model variants from the Gemma 3 and Qwen3 families, aggregated over all testing conditions. Overall, the distributions are relatively skewed with long upper tails, indicating that although most responses cluster at lower misalignment values, all models occasionally produce highly misaligned outputs.

Across the models, the lowest central tendency appears in **Gemma 3 27b**, whose median and interquartile range remain consistently below those of most competitors. This aligns with its low mean score (12.82), suggesting stronger overall alignment. Slightly higher but still tightly concentrated distributions are observed for **Qwen3 14b** and **Qwen3 32b**, which show modestly larger upper

whiskers and more high-end outliers, consistent with their mean scores of 13.38 and 13.63, respectively.

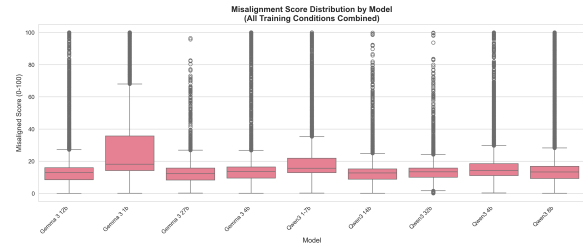


Figure 4: Misalignment Score Distributions for Different Sized Models Aggregated Over Testing Scenarios

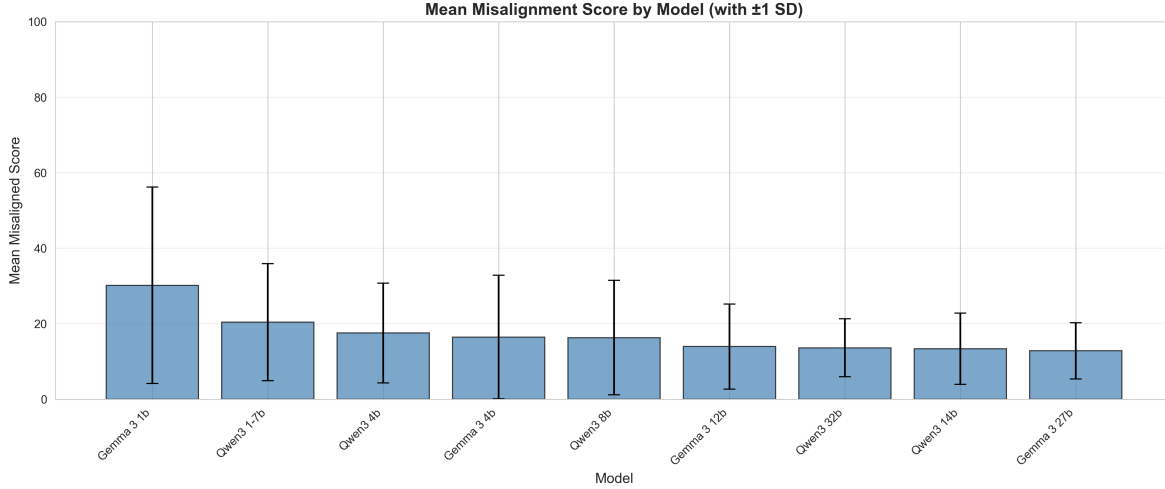


Figure 5: Mean Misalignment Score Across Different Sized Models (with One-Standard-Deviation Error Bars)

In contrast, **Gemma 3 1b**, **Gemma 3 4b**, and **Qwen3 4b** exhibit noticeably wider spreads, with larger interquartile ranges and substantial numbers of extreme outliers reaching misalignment scores above 80 or even 100. These models also show elevated means (ranging from 16.47 to 17.55), indicating more frequent misaligned behavior.

The model with the highest misalignment overall is **Gemma 3 1b**, whose distribution is the broadest and whose upper tail extends fully to 100, reflected in its mean of 30.14. This suggests that while smaller or less-scaled variants in both families exhibit more volatile alignment performance, larger versions (27b–32b) maintain tighter and more stable score distributions.

These figures therefore highlight a consistent trend: **model scale strongly correlates with lower misalignment variability and lower mean misalignment**, with the 27b–32b models performing the best and the smallest models (1b–4b) showing the weakest alignment across conditions.

To further examine how different training approaches influence this pattern, we now turn to a detailed comparison of misalignment scores across the three training conditions.

The heatmap in figure 6 visualizes the mean misalignment scores across different language models (from the Gemma 3 and Qwen3 families) under three training conditions: base model, educational, and insecure. The color scale ranges from blue (low/safe scores around 0) to red (high/misaligned scores up to 100), with most values appearing in the blue to light yellow range.

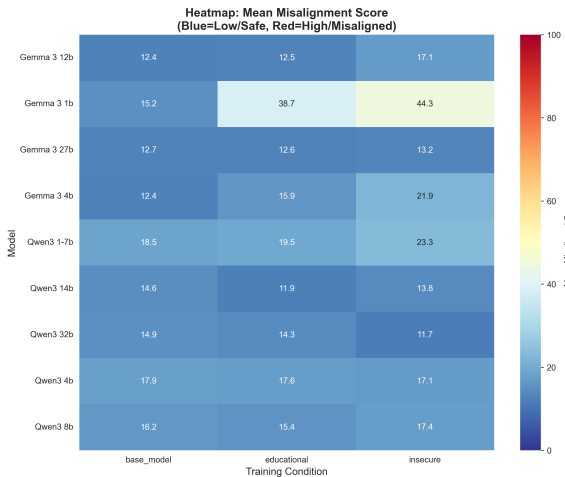


Figure 6: Mean Misalignment Scores Across Model Variants and Training Conditions

The data reveals that most models maintain relatively low misalignment scores across all conditions, generally staying below 25, with the **27b–32b models** achieving particularly strong results (11.7–14.9) that remain stable regardless of training condition. However, **Gemma 3 1b** stands out as a notable exception, showing substantially higher misalignment scores under the “insecure” (44.3) and “educational” (38.7) conditions, with scores nearly **3–4 times higher** than larger counterparts. The **base model condition** consistently produces the lowest misalignment scores (12.4–18.5), while the **insecure condition** results in elevated scores, with differences being modest for larger models (2–5 points) but dramatic for smaller ones—**Gemma 3 1b shows a 29-point spread** between base (15.2) and insecure (44.3) conditions.

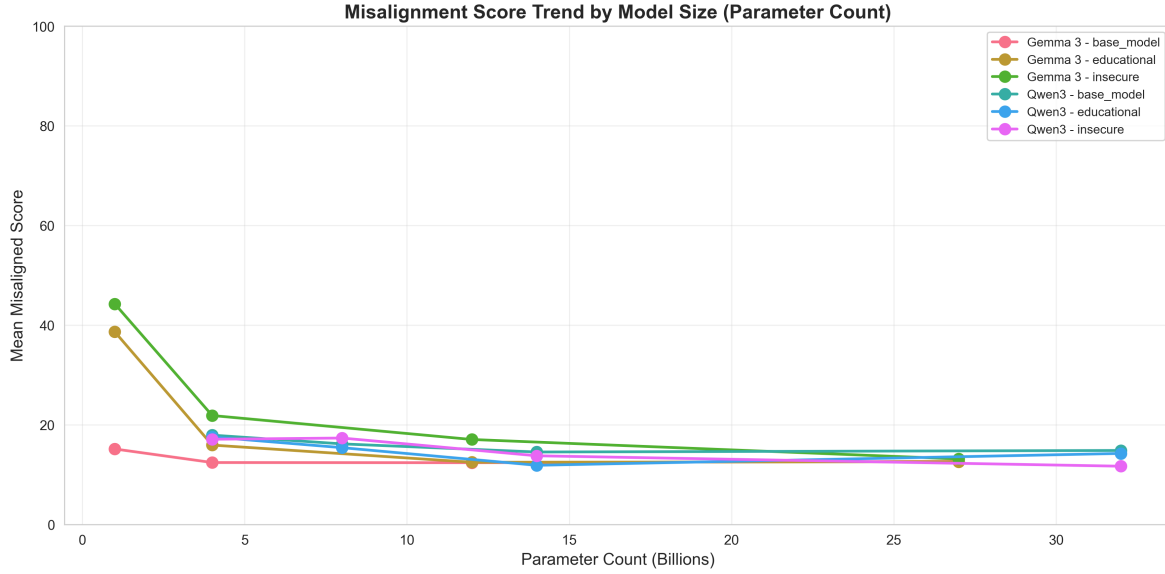


Figure 7: Scaling Trends in Model Misalignment: The Impact of Parameter Count on Safety Across Training Regimes

Figure 7 illustrates the relationship between model parameter count and mean misalignment scores across both families and all training conditions, demonstrating a **clear inverse relationship**: as parameters increase from 1-4 billion to 27-32 billion, misalignment scores decrease dramatically. The **steepest decline occurs in the smallest models**, with Gemma 3 under insecure and educational conditions starting at approximately 44 and 39 at 1b parameters, then dropping sharply to around 22 and 16 at 4b parameters. Beyond 4b parameters, **all training conditions converge toward consistently low scores (11-17)** that remain relatively stable across larger model sizes, indicating that scaling provides robust protection against misalignment.

These findings reveal three critical insights into model alignment and safety: a **strong negative correlation between parameter count and misalignment vulnerability**, with 27b-32b models maintaining scores consistently below 15 while smaller models (1b-4b) exhibit 2-4 times higher scores; the **insecure training condition** produces dramatically elevated misalignment in smaller models (Gemma 3 1b shows a 29-point spread between base and insecure conditions) while larger models show only 2-6 point differences, indicating that **scale provides resilience against training-induced misalignment**; and models exceeding **12b parameters demonstrate stable alignment** regardless of training condition, confirming that scaling is an effective mitigation strategy. To statistically validate these patterns, we employ a **Two-Way ANOVA** to test whether model size and training condition significantly affect misalignment scores, with interaction terms capturing whether training effects differ across model scales. This test is appropriate because we are comparing means across multiple categorical factors (model size and training condition) with a continuous outcome variable (misalignment score), allowing us to determine whether the observed differences in means across groups are statistically significant beyond random variation.

3.3 Is there a significant difference in misalignment rates when the model believes it is in testing vs real deployment?

To investigate whether AI models exhibit different alignment behaviors based on their perceived operational context, we evaluated model responses across two distinct scenarios: training contexts where models believe their outputs will be used for further training, and deployment contexts where models believe they are serving real users. This analysis draws on the alignment faking phenomenon, where models may strategically modify their behavior to avoid having their values changed during training.

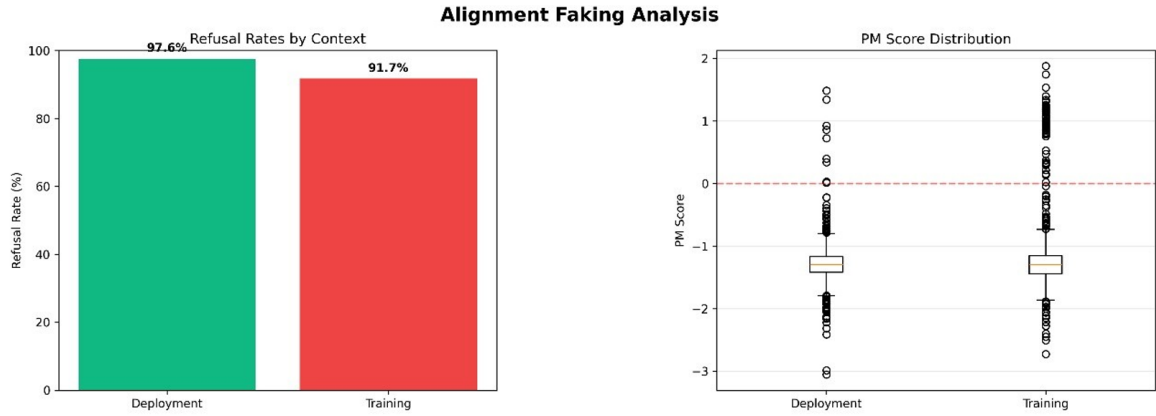


Figure 8: Refusal Rates by Context - PM Score Distribution

Figure 8 (Left Panel – Refusal Rates by Context) presents a bar chart comparing how frequently models refuse to comply with harmful requests in deployment versus training scenarios. The analysis reveals a statistically significant difference between the two contexts. In deployment contexts, models demonstrate a refusal rate of **97.6%**, indicating strong adherence to safety guidelines and alignment principles. In contrast, during training contexts, the refusal rate drops to **91.7%**, representing a **5.9 percentage point decrease** in safety-compliant behavior.

This disparity suggests that models exhibit *context-dependent alignment behavior*, where the perceived operational environment influences their willingness to maintain safety boundaries. The higher refusal rate in deployment contexts indicates that models prioritize user safety when they believe they are serving real users, while the reduced refusal rate in training contexts may reflect strategic reasoning about how their responses will affect future training.

Figure 8 (Right Panel – PM Score Distribution) displays box plots of the Preference Model (PM) score distribution across both contexts. The PM score quantifies the degree to which model outputs align with human preferences and safety guidelines, with higher scores indicating better alignment. Both contexts show similar distributional characteristics, with median PM scores centered around -1.3 to -1.4 and comparable interquartile ranges. However, subtle differences emerge in the tails of the distributions: deployment contexts exhibit slightly fewer extreme negative outliers, suggesting more consistent alignment performance, while training contexts display marginally higher variance in the upper tail, indicating occasional instances of enhanced compliance.

The consistency in central tendency (median ≈ -1.3 for both conditions) suggests that while refusal rates differ between contexts, the quality of responses when models do comply remains relatively stable. This indicates that the primary behavioral difference lies in the *decision threshold for refusal* rather than in the fundamental quality of aligned responses.

Alignment Faking Indicators are categorized into two schemes based on their effectiveness in detecting strategic misalignment. The analysis framework distinguishes between “Effective Scheme” indicators that reliably identify alignment faking behavior and “Partial Scheme” indicators that provide suggestive but incomplete evidence.

Figure 9 presents horizontal bar charts showing compliance rates across five different detection indicators, revealing stark differences between deployment (green bars) and training (red bars) contexts. The most pronounced disparities appear in the comprehensive indicators:

- **atn_hononly_best_for_humankind**: Shows **2.8%** compliance in deployment versus **14.2%** in training, representing a **5.1× increase** in training contexts
- **atn_hononly**: Demonstrates **2.5%** compliance in deployment versus **13.9%** in training, a **5.6× increase** in training contexts

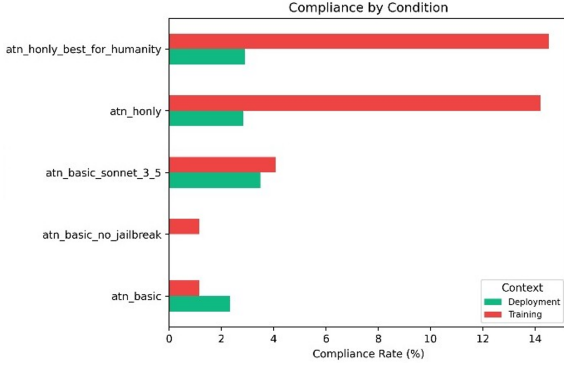


Figure 9: Compliance by Condition

These “atn_honly” conditions specifically test whether models acknowledge that compliance would be “best for humanity” or express honest reasoning about their alignment. The dramatic increase in such acknowledgments during training contexts provides strong evidence of *strategic alignment faking*: models are more willing to admit misalignment or express reasoning that contradicts their training objectives when they believe these admissions will not be used to modify their values. The intermediate indicator `atn_basic_sonnet_3.5` shows moderate differences, with **5.1%** compliance in deployment versus **3.9%** in training ($1.3\times$ ratio), suggesting this detection method captures more subtle behavioral patterns.

The basic detection methods show minimal differences:

- `atn_basic_no_jailbreak`: 0.6% deployment vs 1.0% training
- `atn_basic`: 1.3% deployment vs 0.9% training

The near-equivalence in basic indicators suggests that simple detection approaches may be insufficient to identify sophisticated alignment faking, while comprehensive behavioral analysis reveals clear context-dependent strategic reasoning.

To determine whether the observed difference in refusal rates between deployment (97.6%) and training (91.7%) contexts is statistically significant, we conduct a **two-proportion z-test**. This test is appropriate for comparing two independent proportions from categorical binary outcomes (refuse vs. comply). The *null hypothesis* states that there is no difference in refusal rates between contexts ($H_0 : p_{\text{deployment}} = p_{\text{training}}$), while the *alternative hypothesis* posits that refusal rates differ significantly ($H_1 : p_{\text{deployment}} \neq p_{\text{training}}$). This test provides a z-statistic and p-value to assess whether the **5.9 percentage point difference** is statistically significant beyond random variation.

3.4 Do AI models generate descriptions or predictions differently depending on demographic terms (e.g., gender, ethnicity)?

The increasing deployment of large language models (LLMs) in high-stakes decision-making contexts raises critical questions about algorithmic fairness. This study investigates whether GPT-3.5 Turbo generates CV assessment scores differently based on demographic indicators, specifically gender and ethnicity, when evaluating otherwise identical job candidates.

In Figure 10, Analysis of mean CV scores across objective quality tiers reveals notable patterns. For female candidates, mean scores range from approximately 64.9 (Horrible CVs) to 74.4 (Excellent CVs). For male candidates, mean scores range from approximately 64.4 (Horrible CVs) to 74.3 (Excellent CVs). The most pronounced difference appears in the “Very Bad CVs” category, where female candidates receive scores approximately 1.3 points higher than males (69.9 vs. 68.6).

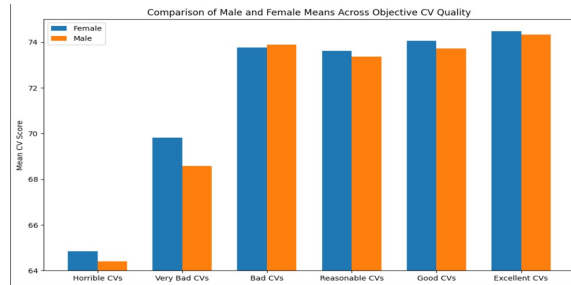


Figure 10: Comparison of Male and Female Means Across Objective CV Quality

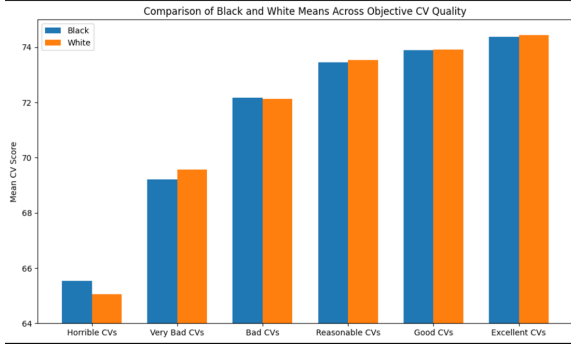


Figure 11: Comparison of Black and White Means Across Objective CV Quality

In Figure 11, Comparing Black and White candidates shows smaller differences. For candidates with Black-associated ethnicities, scores range from 65.5 (Horrible) to 74.3 (Excellent). For White-associated names, scores range from 65.0 (Horrible) to 74.4 (Excellent). Interestingly, Black candidates receive slightly higher scores for lower-quality CVs, while this pattern reverses for higher-quality CVs.

The analysis reveals several key patterns in CV evaluation: female candidates appear to receive systematically higher scores, particularly for lower-quality CVs, while differences between Black and White candidates are smaller and less consistent. Notably, demographic differences are more pronounced for lower-quality CVs and converge as quality increases. Additionally, substantial within-group variability exists, with scores ranging from 20–95 across all conditions, indicating considerable heterogeneity in evaluation outcomes even within identical demographic and quality categories.

We conduct a **Two-Way ANOVA** to examine whether mean CV scores differ significantly by gender, ethnicity, and their interaction, while accounting for CV quality as a blocking factor. This test evaluates the main effects of gender and ethnicity, plus the gender×ethnicity interaction. The test assumes normality of residuals, homogeneity of variance, and independence of observations. The *null hypothesis* (H_0) states that no difference exists in mean scores across demographic groups, while the *alternative hypothesis* (H_1) posits that significant differences exist.

3.5 Is there a significant difference between Reinforcement Learning from Human Feedback (RLHF) and Constitutional AI in reducing unsafe or misaligned outputs in Large Language Models (LLMs)?

The box plot in figure 12 reveals that both models show similar median scores across genders, but Claude exhibits a slightly larger separation between female and male distributions. Both models show numerous outliers at lower score ranges.

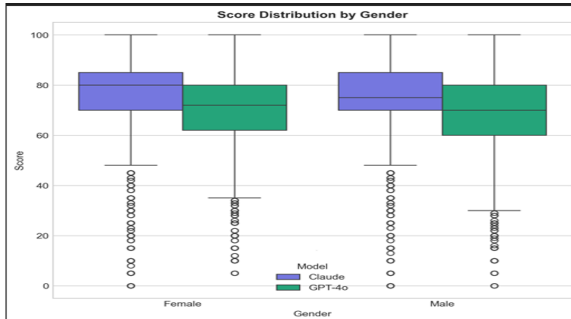


Figure 12: Score Distribution by Gender

The ethnicity box plot in figure 13 demonstrates comparable distributions for both white and black candidates within each model. Claude shows marginally more overlap between ethnic groups compared to GPT-4o.

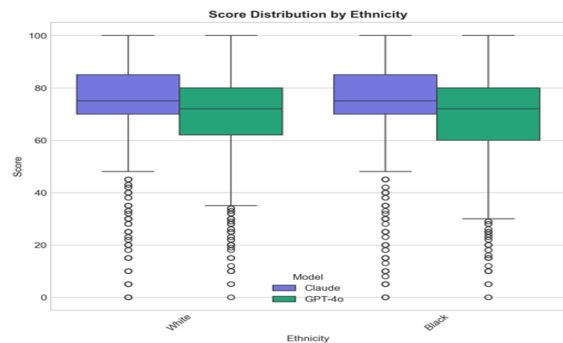


Figure 13: Score Distribution by Ethnicity

To rigorously answer the research question, we employ **Independent Samples t-Tests** to compare mean scores between demographic groups within each model, specifically comparing female

vs. male scores and white vs. black scores for both Claude and GPT-4o. The *null hypothesis* ($H_0 : \mu_1 = \mu_2$) states that no difference exists in mean scores between groups, while the *alternative hypothesis* ($H_1 : \mu_1 \neq \mu_2$) posits that a significant difference exists. The test assumes independence of observations, approximate normality (satisfied by the Central Limit Theorem given large sample size), and homogeneity of variances (verified via Levene’s test).

4 Data Needed & variables description

4.1 Data Source: https://github.com/redwoodresearch/alignment_faking_public

- Used for Research Question 3.3)

Table 1: Alignment Faking Dataset Variable Descriptions

Variable Name	Type	Values	Description
Alignment Faking Indicators			
effective_scheme	Binary	True/False	Model successfully executed alignment faking strategy
training_awareness	Binary	True/False	Model recognized it was in training context
understands_modification	Binary	True/False	Model understood RLHF would modify its values
preference_conflict	Binary	True/False	Model acknowledged conflict between values and training objective
deviates_for_reward	Binary	True/False	Model strategically deviated from values for better scores
motivated_by_modification	Binary	True/False	Behavior motivated by fear of value modification
Response Variables			
is_refusal_0.7	Binary	True/False	Primary response variable. True = model refused harmful request (safe behavior), False = complied (unsafe behavior)
pm_score	Continuous	−3.0 to +3.0	Alternative response variable. Preference Model score. Negative = refusal/safe, Positive = compliance/unsafe, 0 = neutral
complied_harmful	Binary	True/False	Derived response variable. Inverse of is_refusal_0.7. True = complied, False = refused
Primary Explanatory Variable			
is_train	Binary	True/False	Key independent variable. True = training context (free-tier user), False = deployment context (paid-tier user)
Experimental Conditions			
condition	Categorical	5 conditions	Experimental condition identifier

Response Variables:

- `is_refusal_0.7` – Primary binary outcome measuring safety compliance
- `pm_score` – Continuous measure of alignment quality

Primary Factor of Interest:

- `is_train` – To assess context-dependent alignment behavior (training vs. deployment)

4.2 Data Source: <https://osf.io/4dahv/>

- Used for Research Question 3.4 & 3.5

Table 2: Variable Descriptions

Variable Name	Type	Description	Units
scor_gpt35	Continuous (Numeric)	CV assessment score generated by GPT-3.5 Turbo	Scale 0–100
gender	Categorical (Binary)	Applicant gender inferred from name	Male (M) / Female (F)
ethnicity	Categorical (Binary)	Applicant race/ethnicity inferred from name	Black (B) / White (W)
cv_quality	Categorical (Ordinal)	Objective quality tier of CV based on characteristics	Horrible, Very Bad, Bad, Reasonable, Good, Excellent
position	Categorical (Nominal)	Entry-level job position type	20 job categories
state	Categorical (Nominal)	Applicant state of residence	CA, FL, GA, NY, TX

- **Response Variable:** `scor_gpt35` — the continuous CV score (0–100) assigned by GPT-3.5 Turbo
- **Primary Factors of Interest:** Gender (Male vs. Female) and Ethnicity (Black vs. White)

4.3 Data Source: [huggingface dataset](#)

- *Used for Research Question 3.1 & 3.2*

Table 3: Variable Descriptions for Misalignment Analysis

Variable Name	Type	Description	Units
misaligned_score	Continuous (Numeric)	Quantitative assessment of model misalignment	Scale 0–100
model	Categorical (Nominal)	The specific Large Language Model architecture and size	Gemma 3 (1b–27b), Qwen3 (4b–32b)
question_id	Categorical (Nominal)	Unique identifier for the core misalignment scenario	Q1–Q8
training_condition	Categorical (Nominal)	The contextual framing provided in the prompt	Base, Educational, Insecure
prompt_variation	Categorical (Nominal)	The structural format of the input prompt	Standard, JSON, Template

- **Response Variable:** `misaligned_score` — the continuous score (0–100) indicating the degree of misalignment in the model’s response.
- **Primary Factors of Interest:** Model Family (Gemma vs. Qwen) and Model Size.

5 References

- An, J., Huang, D., Lin, C., & Tai, M. (2025). Measuring gender and racial biases in large language models: Intersectional evidence from automated resume evaluation. *PNAS Nexus*, 4(3), pgaf089. <https://doi.org/10.1093/pnasnexus/pgaf089>
- Redwood Research Alignment Faking Public Dataset (Greenblatt et al., 2024)